

Control-Effort-Based Robust Weapon-Target Assignment Under Predicted Intercept Point Uncertainty

Journal Title
XX(X):1–16
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DOI: 10.1177/ToBeAssigned
www.sagepub.com/

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Abstract

Many-to-many missile defense engagements require assignment decisions that remain effective under uncertainty in predicted intercept points (PIPs). This paper presents a robust Weapon-Target Assignment (WTA) framework in which pairwise assignment costs are generated from an assignment-level reduced-order planar linear optimal-control surrogate that quantifies the lateral diversion effort required for an interceptor to reach a candidate PIP. Candidate PIP realizations are generated either by random sampling or by deterministic sigma points, and the resulting cost vectors are enclosed by an ellipsoidal uncertainty set constructed using the Minimum Volume Enclosing Ellipsoid (MVEE). The deterministic binary assignment problem is then reformulated as an integer second-order cone program (SOCP) that hedges against worst-case cost realizations within the modeled uncertainty set. Monte Carlo simulations in 6-vs-4 engagement scenarios are used to compare the proposed approach with deterministic optimization, a budgeted-uncertainty combinatorial baseline, and an exhaustive sample-based brute-force benchmark. The proposed method reduces evaluation-set worst-case degradation relative to deterministic optimization and the budgeted-uncertainty combinatorial baseline. The random-sampling variant provides a more conservative empirical protection, whereas the sigma-point variant offers a more favorable robustness–runtime trade-off with substantially lower computation time in the scenarios studied. Additional scalability, parameter-sensitivity, robust-objective calibration, and PN-based three-dimensional validation studies are conducted to assess computational scaling, uncertainty-set conservatism, and assignment-level ranking consistency of the reduced-order surrogate.

Keywords

Weapon-Target Assignment, Predicted Intercept Point Uncertainty, Ellipsoidal Uncertainty Set, Robust Optimization, Second-Order Cone Programming

Introduction

Missile defense engagements are increasingly characterized by many-to-many interaction patterns, in which multiple interceptors must be allocated to multiple incoming threats under severe time and resource constraints. Within this setting, the Weapon-Target Assignment (WTA) problem plays a central role because the quality of the assignment directly influences overall interception effectiveness^{1–5}. As offensive systems become faster, more maneuverable, and more numerous, assignment strategies based solely on nominal target information become increasingly fragile.

In high-altitude or exoatmospheric engagements, aerodynamic control authority is limited, and interceptors rely heavily on divert and attitude control systems (DACS) to generate lateral maneuvering capability. Because DACS operation is constrained by finite propellant and impulse availability, excessive maneuvering during the midcourse phase can reduce the control margin available for terminal homing and thereby degrade engagement success⁶. This makes control effort a meaningful quantity for assignment. In particular, an assignment strategy that reduces the lateral diversion effort required to reach candidate intercept locations can help preserve terminal maneuvering capability across the engagement set.

A practical difficulty is that many-to-many assignment requires repeated evaluation of pairwise interceptor–target

costs inside a combinatorial optimization loop. Although high-fidelity nonlinear three-dimensional engagement models are valuable for detailed simulation and guidance analysis, embedding them directly in a large-scale assignment routine can be computationally burdensome. The objective of this paper is therefore to develop an assignment-level cost generator that remains physically interpretable while being computationally tractable for repeated use inside the WTA loop. To this end, we adopt a reduced-order planar linear optimal-control surrogate that estimates the lateral diversion effort required for an interceptor to reach a target's predicted intercept point (PIP).

Although a broad literature exists on WTA, most formulations assume that target information is known accurately

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at the time of assignment. Existing studies have emphasized exact or approximate solution methods, constructive heuristics, metaheuristics, and decentralized implementations^{4,5,7-9}. More generic robust discrete optimization frameworks, such as budgeted uncertainty models, provide useful protection against coefficient perturbations¹⁰. However, in missile defense applications the uncertainty often originates upstream in the PIP itself, and this uncertainty induces correlated changes across multiple pairwise assignment costs. Such structure is not naturally represented by nominal deterministic formulations and may be only coarsely captured by uncertainty models that treat cost perturbations as largely independent.

Motivated by this gap, this paper develops a robust WTA framework that propagates PIP uncertainty into the assignment-cost space and then optimizes against the resulting worst-case cost realizations within a modeled uncertainty set. Candidate PIP realizations are generated either by random sampling or by deterministic sigma points, the corresponding cost vectors are computed using the assignment-level control-effort model, and the resulting point cloud is enclosed by an ellipsoidal set constructed using the Minimum Volume Enclosing Ellipsoid (MVEE)^{11,12}. The robust counterpart of the deterministic binary assignment problem is then formulated as an integer second-order cone program (SOCP). The contribution of this paper lies in integrating sampled PIP uncertainty propagation, ellipsoidal cost modeling, and mixed-integer conic optimization into a robust WTA framework driven by a control-effort-based pairwise cost model. Once the sampled cost cloud has been compressed into the ellipsoid parameters (E, c) , the final assignment stage no longer depends on the number of sampled PIP realizations.

An important feature of the framework is its modularity. Although this paper uses a reduced-order planar surrogate to preserve tractability and isolate the effect of PIP uncertainty on lateral diversion effort, the outer robust-assignment layer is not tied to this particular cost generator. The same pipeline can therefore be combined with higher-fidelity simulations, precomputed engagement tables, or surrogate models of different fidelities, provided that they generate pairwise assignment costs for each PIP realization.

The main contributions of this paper are summarized as follows:

1. We formulate a robust WTA problem in which uncertainty in the PIPs is mapped to uncertainty in the assignment cost vector, and we show that the resulting robust counterpart of the deterministic binary assignment problem can be written as an integer SOCP.
2. We introduce a sampling-based uncertainty-construction procedure that uses either random PIP realizations or deterministic sigma points, thereby exposing a practically relevant trade-off between empirical robustness and computational cost.
3. We employ a control-effort-based assignment metric derived from a reduced-order linear optimal-control surrogate, which yields physically interpretable pairwise costs tied to lateral diversion effort rather than generic geometric distance alone.

4. We provide Monte Carlo comparisons, parameter-sensitivity analyses, scalability tests, and a dynamics-based validation against a three-dimensional PN simulation, thereby clarifying the robustness, computational cost, and practical operating regime of each variant of the proposed framework.

The numerical results reported later show that the proposed framework improves upper-tail and worst-case performance relative to deterministic and budgeted-uncertainty baselines in the 6-vs-4 engagement scenarios studied. They also show that the two proposed variants occupy different operating points on a robustness-runtime spectrum: the random-sampling variant provides more conservative empirical protection, whereas the sigma-point variant substantially reduces uncertainty-set construction cost. These results suggest that the framework can support periodically updated assignment decisions under modeled PIP uncertainty when the fidelity of the cost generator, the uncertainty-set construction method, and the computational budget are chosen consistently.

The remainder of this paper is organized as follows. First, we formulate the deterministic WTA problem and introduce the assignment-level control-effort model. Next, we present the robust optimization framework and the MVEE-based uncertainty-set construction. Following this, we report the simulation setup and numerical results. Finally, we conclude the paper and discuss limitations and future research directions.

Problem Statement and Deterministic WTA Formulation

This section defines the deterministic assignment problem and the pairwise cost model used throughout the paper. For each PIP realization later used to construct uncertainty sets and to evaluate robustness in the Monte Carlo study, the pairwise cost matrix is computed from the control-effort model developed here.

Assignment-Level Control-Effort Surrogate

The WTA problem considered in this paper seeks an assignment that minimizes the total lateral diversion effort required for the interceptors to reach the PIPs of the assigned targets. Quantities such as zero-effort miss and terminal heading error are closely related to the lateral acceleration required during the engagement. Using control effort as the assignment cost is therefore physically interpretable and operationally meaningful¹³⁻¹⁵.

The dynamic model introduced below serves as an assignment-level reduced-order surrogate for pairwise cost generation. It captures the dominant transverse diversion effort over short midcourse horizons while preserving the computational structure required for repeated evaluation inside the assignment loop. The surrogate is motivated by short midcourse assignment horizons in high-altitude or exoatmospheric engagements, in which the interceptor already possesses a large forward speed along its nominal flight direction.

In the formulation, the engagement geometry is generated in three-dimensional space, and the planar surrogate is

applied only after transforming each interceptor–target pair into an interceptor-dependent local transverse plane. Let $\mathbf{r}_m^{(j)} \in \mathbb{R}^3$ and $\mathbf{v}_m^{(j)} \in \mathbb{R}^3$ denote the position and velocity of interceptor j , respectively, and let $\mathbf{p}^{(i)} \in \mathbb{R}^3$ denote a PIP realization of target i . For each interceptor, the local longitudinal direction is defined by its velocity vector as

$$\mathbf{e}_z^{(j)} = \frac{\mathbf{v}_m^{(j)}}{\|\mathbf{v}_m^{(j)}\|_2}. \quad (1)$$

Two orthonormal transverse directions, $\mathbf{e}_x^{(j)}$ and $\mathbf{e}_y^{(j)}$, are selected such that $\mathbf{e}_x^{(j)} \times \mathbf{e}_y^{(j)} = \mathbf{e}_z^{(j)}$. The corresponding rotation matrix is

$$R_j = \begin{bmatrix} (\mathbf{e}_x^{(j)})^T \\ (\mathbf{e}_y^{(j)})^T \\ (\mathbf{e}_z^{(j)})^T \end{bmatrix}. \quad (2)$$

The three-dimensional relative PIP position is projected into this local frame as

$$\boldsymbol{\rho}_{ij} = R_j \left(\mathbf{p}^{(i)} - \mathbf{r}_m^{(j)} \right) = [\rho_{ij,x} \quad \rho_{ij,y} \quad \rho_{ij,z}]^T. \quad (3)$$

Only the transverse components are used in the assignment-level control-effort surrogate,

$$y_{\text{des},ij} = [\rho_{ij,x} \quad \rho_{ij,y}]^T. \quad (4)$$

Thus, the proposed cost model does not assume that the entire engagement is intrinsically two-dimensional. Instead, each pairwise cost is evaluated on the local transverse plane normal to the current interceptor velocity, while the original PIP uncertainty and engagement geometry remain three-dimensional. The downrange component $\rho_{ij,z}$ is treated as being governed primarily by the interceptor's existing forward motion over the assignment horizon; the assignment cost therefore measures the lateral diversion effort required to steer toward the projected PIP. The outer robust WTA formulation developed later is agnostic to the specific pairwise cost generator; in principle, higher-fidelity engagement models or precomputed cost tables could be substituted within the same framework.

Under these assumptions, the transverse motion of each interceptor in its local plane is approximated by the standard planar double-integrator model

$$\dot{\mathbf{p}}_{\perp}(t) = \mathbf{v}_{\perp}(t), \quad \dot{\mathbf{v}}_{\perp}(t) = \mathbf{u}_{\perp}(t), \quad (5)$$

where $\mathbf{u}_{\perp}(t) = [a_x(t) \ a_y(t)]^T \in \mathbb{R}^2$ denotes the commanded transverse acceleration generated by the divert and attitude control system, and $\mathbf{v}_{\perp}(t) = [v_x(t) \ v_y(t)]^T \in \mathbb{R}^2$ and $\mathbf{p}_{\perp}(t) = [p_x(t) \ p_y(t)]^T \in \mathbb{R}^2$ denote the local transverse velocity and position, respectively. For high-altitude or exoatmospheric interception scenarios, aerodynamic damping in the transverse motion is negligible. Therefore, the PIP-to-interceptor lateral motion is modeled as a standard planar double-integrator system without a damping term.

Using a first-order discrete-time approximation with sampling interval Δt , the surrogate is written as

$$x_{k+1} = Ax_k + Bu_k, \quad (6)$$

$$y_k = Cx_k, \quad (7)$$

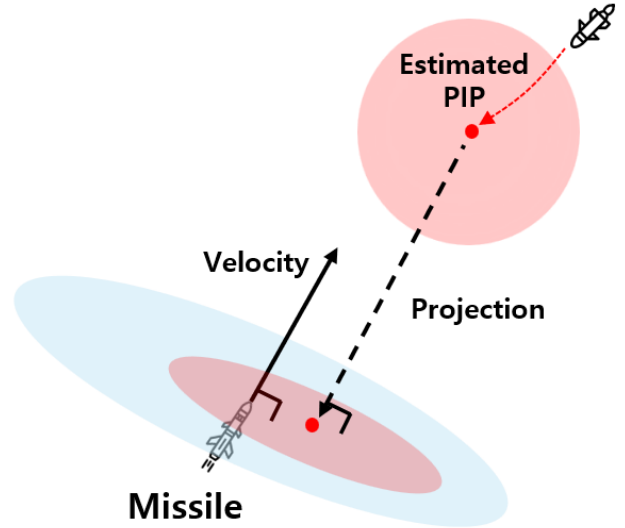


Figure 1. Illustration of the three-dimensional engagement geometry and interceptor-dependent transverse projection used in the assignment-level surrogate. The estimated PIP is defined in the global three-dimensional engagement space, while each interceptor constructs a local frame using its velocity direction as the longitudinal axis. The relative PIP position is then projected onto the local transverse plane, and the projected two-dimensional displacement is used to compute the lateral control-effort-based assignment cost.

where

$$x_k = [p_{k,x} \ p_{k,y} \ v_{k,x} \ v_{k,y}]^T, \quad u_k = [a_{k,x} \ a_{k,y}]^T, \\ y_k = [p_{k,x} \ p_{k,y}]^T.$$

For the double-integrator surrogate used in the revised simulations, the discrete-time matrices are

$$A = \begin{bmatrix} I_2 & \Delta t I_2 \\ 0_2 & I_2 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{2} \Delta t^2 I_2 \\ \Delta t I_2 \end{bmatrix}, \\ C = [I_2 \quad 0_2],$$

where I_2 and 0_2 denote the 2×2 identity and zero matrices, respectively. If a different fixed discrete-time surrogate is used in the numerical implementation, the derivations below remain unchanged and only the specific matrices A , B , and C are replaced accordingly.

Minimum-Norm Pairwise Cost

Let K denote the terminal prediction horizon. By recursively applying (6), the predicted terminal output can be written as

$$y_K = CA^K x_0 + \sum_{k=0}^{K-1} CA^{K-1-k} Bu_k, \quad (8)$$

where A^K denotes the K -step state-transition matrix.

Define the stacked control vector

$$\mathbf{u} = [u_0^T \ u_1^T \ \cdots \ u_{K-1}^T]^T$$

and the output-reachability matrix

$$H_K = [CA^{K-1}B \ CA^{K-2}B \ \cdots \ CB]. \quad (9)$$

Then (8) becomes

$$y_K = CA^K x_0 + H_K \mathbf{u}. \quad (10)$$

Given a desired terminal position y_{des} , the minimum-norm control sequence that drives the surrogate model to that point at step K is

$$\mathbf{u}^* = H_K^T (H_K H_K^T)^{-1} (y_{\text{des}} - CA^K x_0), \quad (11)$$

provided that H_K has full row rank, *i.e.*, the terminal output is reachable at horizon K . The quantity $\|\mathbf{u}^*\|_2^2$ is used as the scalar pairwise assignment cost.

Now consider N targets and M interceptors. For the i -th target and the j -th interceptor, let $y_{\text{des},ij}$ denote the projected PIP in the local transverse plane of interceptor j , as defined in (4), and let $x_0^{(j)}$ denote the corresponding initial transverse state of interceptor j . The corresponding pairwise cost is defined as

$$D_{ij} := \|\mathbf{u}_{ij}^*\|_2^2, \quad i = 1, \dots, N, \quad j = 1, \dots, M, \quad (12)$$

where $\mathbf{u}_{ij}^*$ is obtained from (11) using $x_0 = x_0^{(j)}$ and $y_{\text{des}} = y_{\text{des},ij}$. For every PIP realization later used in the robust formulation and in the Monte Carlo study, the full cost matrix D is generated by repeating this computation for all interceptor–target pairs.

Deterministic WTA Formulation

Let $D \in \mathbb{R}^{N \times M}$ denote the pairwise cost matrix whose entries are given by (12). The deterministic WTA problem assigns interceptors to targets so as to minimize the total pairwise control-energy surrogate. We assume $M \geq N$, each interceptor is assigned to exactly one target, and each target receives at least one interceptor.

Define the binary assignment matrix $Z \in \{0, 1\}^{N \times M}$, where $Z_{ij} = 1$ indicates that interceptor j is assigned to target i . The deterministic WTA problem is then formulated as

$$\begin{aligned} & \underset{Z}{\text{minimize}} && \sum_{i=1}^N \sum_{j=1}^M D_{ij} Z_{ij} \\ & \text{subject to} && \sum_{i=1}^N Z_{ij} = 1, \quad j = 1, \dots, M, \\ & && \sum_{j=1}^M Z_{ij} \geq 1, \quad i = 1, \dots, N, \\ & && Z \in \{0, 1\}^{N \times M}. \end{aligned} \quad (13)$$

Problem (13) is a binary linear assignment problem with a cost matrix induced by the control-effort surrogate. In the deterministic setting, D is computed from nominal PIPs. In practice, however, the PIPs are uncertain, due to sensor limitations and the maneuverability of modern threats such as hypersonic glide vehicles, and therefore the cost matrix itself is uncertain. The next section replaces the single nominal cost matrix by a set-valued uncertainty model in cost space and derives the corresponding robust counterpart.

Robust Optimization Framework for Target Assignment Under Uncertainty

The deterministic WTA problem in (13) assumes that each target is represented by a single nominal PIP. In practice, however, every plausible PIP realization induces a different pairwise cost matrix through (12). The purpose of this section is therefore to replace the single nominal cost matrix by a compact uncertainty description in cost space and to derive the corresponding robust counterpart.

The overall procedure has four stages. First, candidate PIP realizations are generated according to an assumed uncertainty model. Second, for each realization, the corresponding pairwise cost vector is computed using the control-effort surrogate introduced in (11) and (12). Third, the resulting cloud of cost vectors is summarized by an ellipsoidal uncertainty set constructed using a Minimum Volume Enclosing Ellipsoid (MVEE). Fourth, the robust assignment problem induced by that ellipsoid is solved as a mixed-integer second-order cone program (SOCP). This separation gives the framework a clear computational structure: the MVEE stage compresses many sampled cost vectors into a compact ellipsoidal description, after which the final assignment is obtained from a single mixed-integer SOCP.

Ellipsoidal Robust Optimization Formulation

Robust optimization hedges against worst-case realizations of uncertain problem data within a prescribed uncertainty set¹⁶. In the present setting, the uncertainty originates in the PIPs and is mapped to uncertainty in the assignment cost vector.

Let $n_c = NM$ denote the dimension of the vectorized cost space. Define

$$z = \text{vec}(Z) \in \{0, 1\}^{n_c}, \quad d = \text{vec}(D) \in \mathbb{R}^{n_c},$$

where $Z \in \{0, 1\}^{N \times M}$ and $D \in \mathbb{R}^{N \times M}$ are the binary assignment matrix and the pairwise cost matrix defined in (13). For notational clarity, the assignment constraints are still written in terms of Z , while the objective is expressed through its vectorization z . The robust WTA problem is

$$\begin{aligned} & \underset{z}{\text{minimize}} && \max_{d \in \mathcal{U}_d} d^T z \\ & \text{subject to} && \sum_{i=1}^N Z_{ij} = 1, \quad j = 1, \dots, M, \\ & && \sum_{j=1}^M Z_{ij} \geq 1, \quad i = 1, \dots, N, \\ & && z = \text{vec}(Z), \quad z \in \{0, 1\}^{n_c}, \end{aligned} \quad (14)$$

where $\mathcal{U}_d \subset \mathbb{R}^{n_c}$ denotes the set of plausible cost vectors induced by uncertain PIPs.

Assume that $\mathcal{U}_d$ is modeled by an ellipsoid

$$\begin{aligned} \mathcal{U}_d &= \{d \in \mathbb{R}^{n_c} \mid (d - c)^T E (d - c) \leq 1\} \\ &= \{c + E^{-1/2} w \mid \|w\|_2 \leq 1\}, \end{aligned} \quad (15)$$

where $c \in \mathbb{R}^{n_c}$ is the ellipsoid center and $E \in \mathbb{R}^{n_c \times n_c}$ is a symmetric positive definite matrix describing its shape.

Using the representation in (15), the inner maximization in (14) can be evaluated analytically:

$$\begin{aligned} \max_{d \in \mathcal{U}_d} d^T z &= \max_{\|w\|_2 \leq 1} (c + E^{-1/2} w)^T z \\ &= c^T z + \max_{\|w\|_2 \leq 1} w^T E^{-1/2} z \\ &= c^T z + \|E^{-1/2} z\|_2, \end{aligned} \quad (16)$$

where w is an auxiliary vector confined to the unit ℓ_2 -ball. The last equality follows from the support function of the unit Euclidean ball.

Substituting (16) into (14) yields the robust counterpart

$$\begin{aligned} &\underset{z}{\text{minimize}} && c^T z + \|E^{-1/2} z\|_2 \\ &\text{subject to} && \sum_{i=1}^N Z_{ij} = 1, \quad j = 1, \dots, M, \\ & && \sum_{j=1}^M Z_{ij} \geq 1, \quad i = 1, \dots, N, \\ & && z = \text{vec}(Z), \quad z \in \{0, 1\}^{n_c}, \end{aligned} \quad (17)$$

which is a mixed-integer SOCP. An equivalent epigraph form is

$$\begin{aligned} &\underset{z, t}{\text{minimize}} && c^T z + t \\ &\text{subject to} && \|E^{-1/2} z\|_2 \leq t, \\ & && \sum_{i=1}^N Z_{ij} = 1, \quad j = 1, \dots, M, \\ & && \sum_{j=1}^M Z_{ij} \geq 1, \quad i = 1, \dots, N, \\ & && z = \text{vec}(Z), \quad z \in \{0, 1\}^{n_c}. \end{aligned} \quad (18)$$

Problem (18) is the optimization model actually passed to the solver once E and c have been computed. In particular, the final assignment stage no longer depends on the number of sampled PIP realizations m . This clarifies the computational role of the MVEE stage: it compresses many sampled cost vectors into a compact conic description¹⁷, after which the robust WTA problem is solved as a single mixed-integer SOCP using solvers such as GUROBI or ECOS_BB via CVXPY modeling interface.

MVEE-Based Construction of the Uncertainty Set

Let

$$\mathcal{S} = \{d_1, d_2, \dots, d_m\} \subset \mathbb{R}^{n_c}$$

denote a finite set of sampled cost vectors. The goal of the MVEE stage is to compute the minimum-volume ellipsoid that contains all points in $\mathcal{S}$. Specifically, we seek an ellipsoid

$$\mathcal{E}_d(E, c) = \{d \in \mathbb{R}^{n_c} \mid (d - c)^T E (d - c) \leq 1\} \quad (19)$$

with minimum volume subject to $\mathcal{S} \subset \mathcal{E}_d(E, c)$. An example of a two-dimensional MVEE is shown in Fig. 2.

The volume of the ellipsoid $\mathcal{E}_d(E, c)$ in (19) is proportional to the determinant of $E^{-1/2}$ and is given by:

$$\text{Vol}(\mathcal{E}_d) = \frac{\nu_0}{\sqrt{\det(E)}} = \nu_0 \det(E^{-1})^{\frac{1}{2}}, \quad (20)$$

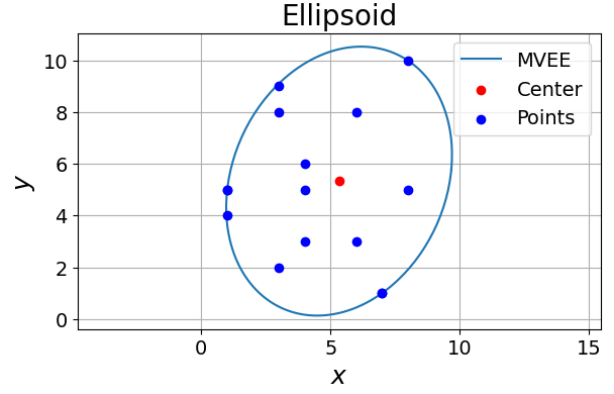


Figure 2. Example of a two-dimensional MVEE.

where ν_0 denotes the volume of the unit n_c -dimensional ball. Thus, the MVEE problem is equivalent to minimizing $\det(E^{-1/2})$ subject to the constraint that all points lie within the ellipsoid:

$$\begin{aligned} &\underset{E, c}{\text{minimize}} && \log \det(E^{-1}) \\ &\text{subject to} && (d_k - c)^T E (d_k - c) \leq 1, \quad k = 1, \dots, m, \\ & && E > 0. \end{aligned} \quad (21)$$

Problem (21) is handled as a preprocessing stage that compresses the sampled set $\mathcal{S}$ into the ellipsoid parameters (E, c) . This preprocessing step decouples uncertainty-set construction from the final robust assignment solve. Direct interior-point solution of (21) can be expensive when the number of samples is large, so we instead use the standard lifted dual formulation of the non-centered MVEE problem^{11,12,18}.

Introduce the lifted vectors

$$\begin{aligned} q_k &= \begin{bmatrix} d_k \\ 1 \end{bmatrix} \in \mathbb{R}^{n_c+1}, \\ Q &= [q_1 \quad q_2 \quad \dots \quad q_m] \in \mathbb{R}^{(n_c+1) \times m}. \end{aligned}$$

Geometrically, the lifted points q_k lie on the hyperplane $z = 1$. By augmenting them with their origin-symmetric reflections $-q_k$ on $z = -1$, one obtains a centrally symmetric lifted set,

$$\mathcal{S}' = \{\pm q_1, \dots, \pm q_m\},$$

whose MVEE is centered at the origin. Figure 3 illustrates this construction schematically. The non-centered ellipsoid in the original cost space is then recovered by intersecting the lifted ellipsoid with the hyperplane $z = 1$.

In the lifted space, the MVEE problem can be written as

$$\begin{aligned} &\underset{S}{\text{minimize}} && \log \det(S^{-1}) \\ &\text{subject to} && q_k^T S q_k \leq 1, \quad k = 1, \dots, m, \\ & && S > 0, \end{aligned} \quad (22)$$

where $S \in \mathbb{R}^{(n_c+1) \times (n_c+1)}$ is the lifted ellipsoid matrix. As illustrated in Figure 3, the lifting converts the non-centered ellipsoid problem in the original cost space into an origin-centered ellipsoid problem over a symmetric point set in the augmented space.

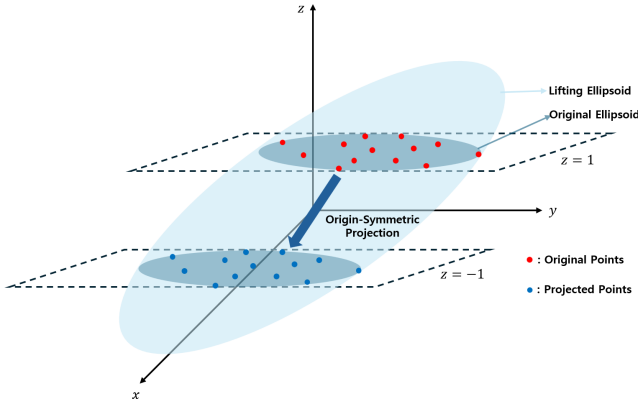


Figure 3. Schematic of the lifted-space construction used for MVEE computation. The original lifted points $q_k = [d_k^T \ 1]^T$ lie on the hyperplane $z = 1$ (red), and their origin-symmetric reflections $-q_k$ lie on $z = -1$ (blue). The shaded ellipsoid denotes the origin-centered MVEE of the symmetric lifted set, and intersecting it with $z = 1$ recovers the non-centered ellipsoid in the original cost space.

Algorithm 1 Standard Frank-Wolfe routine used for MVEE construction

Require: tolerance ε

Ensure: $\lambda \in \mathbb{R}^m$ such that $\mathbf{1}^T \lambda = 1$, $\lambda \geq 0$

1: **Initialization:** $\lambda \leftarrow (1/m)\mathbf{1}$

2: **repeat**

3: **for** $k = 1, \dots, m$ **do**

4: $g_k(\lambda) \leftarrow q_k^T V(\lambda)^{-1} q_k$

5: **end for**

6: $j \leftarrow \arg \max_k g_k(\lambda)$

7: $\kappa \leftarrow \frac{g_j(\lambda) - (n_c + 1)}{(n_c + 1)(g_j(\lambda) - 1)}$

8: $\lambda \leftarrow (1 - \kappa)\lambda + \kappa e_j$

9: **until** convergence criterion is met

Its dual form is

$$\begin{aligned} & \underset{\lambda}{\text{maximize}} && \log \det V(\lambda) \\ & \text{subject to} && \mathbf{1}^T \lambda = 1, \\ & && \lambda \geq 0, \end{aligned} \quad (23)$$

where

$$V(\lambda) = Q \operatorname{diag}(\lambda) Q^T.$$

The dual problem (23) is defined over the unit simplex and can be solved efficiently using a standard Frank–Wolfe routine. This routine is summarized in Algorithm 1. It is included here because it is the computational subroutine used to construct the uncertainty set.

Let $g = [g_1(\lambda) \ \dots \ g_m(\lambda)]^T$ denote the gradient of the dual objective in (23), where each component is given by:

$$g_k(\lambda) = \frac{\partial}{\partial \lambda_k} \log \det V(\lambda) = q_k^T V(\lambda)^{-1} q_k, \quad (24)$$

for $k = 1, \dots, m$. To determine the ascent direction, the index j corresponding to the largest component of the gradient vector g is identified, i.e., $j = \arg \max_i g_i(\lambda)$, and then a closed-form expression for the optimal step size κ is

computed via line search¹⁸:

$$\kappa = \frac{g_j(\lambda) - (n_c + 1)}{(n_c + 1)(g_j(\lambda) - 1)}, \quad (25)$$

which, by construction, satisfies $0 < \kappa < 1$ over the feasible region.

The dual variable is subsequently updated using a convex combination of the current iterate and the selected direction:

$$\lambda^{(t+1)} = (1 - \kappa)\lambda^{(t)} + \kappa e_j.$$

This iterative procedure continues until convergence to an optimal dual solution λ^* . The parameters of the minimum-volume ellipsoid in the original n_c -dimensional space can be recovered as:

$$\begin{aligned} E^* &= \frac{1}{n_c} (P \operatorname{diag}(\lambda^*) P^T - P \lambda^* (P \lambda^*)^T + \epsilon I)^{-1}, \\ c^* &= P \lambda^*, \end{aligned} \quad (26)$$

with

$$P = [d_1 \ \dots \ d_m] \in \mathbb{R}^{n_c \times m}.$$

The diagonal loading term ϵI , with small $\epsilon > 0$, is introduced to ensure numerical regularity when the sampled cost set is low-rank or nearly low-rank, as can occur in the sigma-point construction.

This construction has a direct computational implication. The MVEE stage replaces a potentially large set of sampled cost vectors by the compact parameter pair (E^*, c^*) . After this compression, the robust assignment stage solves only the conic problem in (18), so the final optimization problem does not scale directly with the number of PIP realizations used to construct the uncertainty set.

Propagation of PIP Uncertainty to Cost Space and Robust Assignment

We now describe how uncertainty in the PIPs is propagated into cost space. For the i -th target, the PIP is modeled in the global three-dimensional engagement space as

$$\begin{aligned} \mathbf{p}^i &\sim \mathcal{N}(\bar{\mathbf{p}}^i, \Sigma^i), \\ \Sigma^i &= \operatorname{diag}((\sigma_x^i)^2, (\sigma_y^i)^2, (\sigma_z^i)^2), \end{aligned} \quad (27)$$

where

$$\bar{\mathbf{p}}^i = [\bar{p}_x^i \ \bar{p}_y^i \ \bar{p}_z^i]^T.$$

Let the stacked mean and covariance be

$$\begin{aligned} \bar{\mathbf{p}} &= [(\bar{\mathbf{p}}^1)^T \ \dots \ (\bar{\mathbf{p}}^N)^T]^T \in \mathbb{R}^{3N}, \\ W &= \operatorname{diag}(\Sigma^1, \dots, \Sigma^N) \in \mathbb{R}^{3N \times 3N}. \end{aligned}$$

For every sampled PIP realization $\mathbf{p}_k \in \mathbb{R}^{3N}$, each target PIP is first projected into the interceptor-dependent local transverse plane defined in (3)–(4). The corresponding pairwise cost matrix $D(\mathbf{p}_k)$ is then computed by evaluating (12) for all interceptor–target pairs. The associated cost vector is defined as

$$d_k = \operatorname{vec}(D(\mathbf{p}_k)) \in \mathbb{R}^{n_c}, \quad k = 1, \dots, m.$$

The sampled set passed to the MVEE stage is therefore

$$\mathcal{S} = \{d_1, d_2, \dots, d_m\}.$$

In this paper, the following two sampling strategies are considered.

Random sampling. The candidate PIP realizations are drawn directly from the joint Gaussian model,

$$\mathbf{p}_k \sim \mathcal{N}(\bar{\mathbf{p}}, W), \quad k = 1, \dots, m.$$

This strategy can provide a rich empirical description of the uncertainty, but its computational cost grows with the number of samples m .

Sigma-point sampling. To reduce the number of required realizations, deterministic sigma points inspired by the unscented transform are used^{19,20}. Let $L \in \mathbb{R}^{3N \times 3N}$ satisfy

$$LL^T = \gamma_{\text{sp}} W,$$

where $\gamma_{\text{sp}} > 0$ is a scaling parameter, and let L_i denote the i -th column of L . The sigma points are

$$\begin{aligned} \mathbf{p}_0 &= \bar{\mathbf{p}}, \\ \mathbf{p}_i &= \bar{\mathbf{p}} + L_i, \quad i = 1, \dots, 3N, \\ \mathbf{p}_{3N+i} &= \bar{\mathbf{p}} - L_i, \quad i = 1, \dots, 3N. \end{aligned} \quad (28)$$

Hence, this construction uses a total of $6N + 1$ deterministic realizations. The sigma-point variant therefore produces a much smaller set $\mathcal{S}$ than large-scale random sampling, which is the main source of its computational advantage.

Once $\mathcal{S}$ has been obtained by either sampling strategy, the MVEE problem (21) is solved to obtain (E, c) . The ellipsoidal uncertainty set

$$\mathcal{U}_d = \{d \in \mathbb{R}^{n_c} \mid (d - c)^T E (d - c) \leq 1\}$$

is then substituted into (18), and the resulting mixed-integer SOCP is solved to obtain the robust assignment. This problem is then solved as an integer SOCP using commercial solvers, resulting in an optimal assignment matrix Z^* that minimizes the worst-case total control effort while satisfying all assignment constraints.

The ellipsoidal uncertainty set robustifies the assignment against the modeled range of PIP variations represented by the sampled set. The sigma-point scale γ_{sp} can be increased to enlarge the uncertainty envelope and obtain a more conservative assignment when broader maneuver ranges must be covered.

The full procedure is summarized in Algorithm 2, which makes the computational structure explicit. The sampling stage determines how many cost vectors are evaluated, the MVEE stage compresses those vectors into (E, c) , and the final assignment stage solves a single mixed-integer SOCP. The resulting robustness–runtime trade-off is therefore governed primarily by how the uncertainty set is constructed.

The proposed method delivers robustness with respect to the PIP-induced cost variations represented by the sampled set and its ellipsoidal approximation. Extensions to structural model mismatch and broader uncertainty classes are discussed in the subsection entitled *Discussion of Modeling and Evaluation Limitations*.

Monte Carlo Evaluation of Robust Assignment Algorithms

Simulation Setup

Monte Carlo simulations were performed to evaluate the robustness and computational cost of the proposed robust

Algorithm 2 Robust WTA pipeline under PIP uncertainty

Require: nominal PIPs $\{\bar{\mathbf{p}}^i, \Sigma^i\}_{i=1}^N$, interceptor initial states $\{x_0^{(j)}\}_{j=1}^M$, prediction horizon K , sampling strategy

- 1: Generate candidate PIP realizations $\{\mathbf{p}_k\}_{k=1}^m$
- 2: **for** $k = 1, \dots, m$ **do**
- 3: Compute $D(\mathbf{p}_k)$ using (12)
- 4: Set $d_k \leftarrow \text{vec}(D(\mathbf{p}_k))$
- 5: **end for**
- 6: Construct (E, c) from $\mathcal{S} = \{d_k\}_{k=1}^m$ using Algorithm 1
- 7: Solve the mixed-integer SOCP (18)
- 8: Recover the optimal assignment matrix Z^* from z^*

WTA framework under uncertain PIPs. All experiments were implemented using the CVXPY modeling interface and solved with the GUROBI optimizer. The simulation and implementation parameters are summarized in Table 1.

Each Monte Carlo trial corresponds to an independently generated engagement geometry. The interceptor positions are sampled uniformly inside a three-dimensional ball of radius 10 km centered at $(0, 0, 0)$ km, while the nominal target PIPs are sampled uniformly inside a three-dimensional ball of radius 10 km centered at $(0, 0, 40)$ km. The interceptor velocity direction is initialized toward the target-center direction with a random perturbation of up to 5° , and the interceptor speed is fixed at 2 km/s.

For each target i , the uncertain PIP is modeled as an isotropic three-dimensional Gaussian distribution,

$$\mathbf{p}^i \sim \mathcal{N}(\bar{\mathbf{p}}^i, \sigma_r^2 I_3), \quad \sigma_r = 6 \text{ km}.$$

For every PIP realization, the global three-dimensional target PIP is projected into the interceptor-dependent local transverse plane, and the pairwise cost matrix is computed using the squared minimum-norm control-effort surrogate defined in (12).

Two variants of the proposed framework are tested. In *SOCP (Random)*, the uncertainty set is constructed from $m_{\text{rand}} = 20,000$ randomly sampled PIP realizations. In *SOCP (Sigma)*, the uncertainty set is constructed from $m_{\text{sp}} = 6N + 1 = 25$ sigma points generated using $\gamma_{\text{sp}} = 6$.

These two methods are compared with the following three baselines:

- *Deterministic Optimization:* Solves the nominal assignment problem using only the cost matrix evaluated at the mean PIPs, $\bar{\mathbf{p}}$. This method does not account for PIP uncertainty and therefore serves as the nominal baseline.
- *Budgeted-Uncertainty Combinatorial Optimization:* Uses the budgeted-uncertainty formulation of Bertsimas and Sim¹⁰. The nominal cost matrix is taken as the componentwise minimum over the random construction samples, and the deviation matrix is defined as the componentwise difference between the maximum and minimum sampled costs.
- *Exhaustive Sample-Based Search:* Exhaustively evaluates all feasible assignment matrices and selects the one with the smallest maximum cost over the random construction samples. This method is used as

Table 1. Simulation and implementation parameters.

Quantity	Value
Number of interceptors M	6
Number of targets N	4
Sampling interval Δt	0.05 s
Prediction horizon time	20 s
Prediction horizon steps K	400
Pairwise cost definition	$D_{ij} = \ \mathbf{u}_{ij}^*\ _2^2$
Interceptor initial-position distribution	Uniform in a 3D ball of radius 10 km centered at (0, 0, 0) km
Nominal PIP distribution	Uniform in a 3D ball of radius 10 km centered at (0, 0, 40) km
Interceptor speed	2 km/s
Initial heading perturbation	Up to 5° from the nominal target-center direction
PIP standard deviation	$\sigma_x^i = \sigma_y^i = \sigma_z^i = 6$ km
Random-sampling size for MVEE construction m_{rand}	20,000
Sigma-point size m_{sp}	$6N + 1 = 25$
Sigma-point scale γ_{sp}	6
MVEE regularization parameter ϵ	1×10^{-5}
Out-of-sample evaluation size m_{eval}	10,000
Number of Monte Carlo engagement geometries N_{MC}	300
Budget parameter Γ in Bertsimas–Sim baseline	6
Frank–Wolfe tolerance	1×10^{-4}
Frank–Wolfe maximum iterations	5,000
Solver / software version	GUROBI 13.0.0 via CVXPY 1.6.0 on Python 3.10.16
Computing platform	CPU: AMD Ryzen 5 5600 6-Core (3.50 GHz), RAM: 32GB

a sample-based reference solution, not as an exact robust optimum over the continuous PIP uncertainty distribution. In the 6-vs-4 scenario, the number of feasible assignments is 1,560, so exhaustive evaluation is computationally manageable.

For each Monte Carlo engagement geometry, every algorithm returns a fixed assignment matrix. The detailed evaluation metrics used to compare the returned assignments are defined in the next subsection.

Evaluation Metrics

For each Monte Carlo engagement geometry, each algorithm returns a fixed assignment vector z . The returned assignment is evaluated on two finite sets of PIP realizations. The first is the construction set used to generate the uncertainty model, and the second is an independent out-of-sample test set generated from the same PIP distribution. The construction-set evaluation indicates how the assignment behaves on the samples used during uncertainty-set construction, whereas the out-of-sample evaluation is used as the primary basis for assessing robustness to unseen PIP realizations.

Let $\mathcal{D} = \{d^{(\nu)}\}_{\nu=1}^m$ denote either the construction-set or out-of-sample evaluation cost vectors in the vectorized cost space. For a fixed assignment vector z , the realized assignment cost for sample ν is

$$J^{(\nu)}(z) = (d^{(\nu)})^T z. \quad (29)$$

The mean and maximum realized assignment costs over $\mathcal{D}$ are defined as

$$\begin{aligned} J_{\text{mean}}(z; \mathcal{D}) &= \frac{1}{m} \sum_{\nu=1}^m J^{(\nu)}(z), \\ J_{\text{max}}(z; \mathcal{D}) &= \max_{\nu=1, \dots, m} J^{(\nu)}(z). \end{aligned} \quad (30)$$

The empirical cumulative distribution function of the realized assignment cost is defined as

$$\hat{F}_z(\eta; \mathcal{D}) = \frac{1}{m} \sum_{\nu=1}^m \mathbb{I} \left[J^{(\nu)}(z) \leq \eta \right]. \quad (31)$$

The empirical 99th-percentile cost is then given by

$$J_{0.99}(z; \mathcal{D}) = \inf \left\{ \eta \in \mathbb{R} \mid \hat{F}_z(\eta; \mathcal{D}) \geq 0.99 \right\}. \quad (32)$$

Thus, $J_{0.99}$ is the empirical cost threshold exceeded by only 1% of the evaluation samples.

The robustness comparison emphasizes J_{max} and $J_{0.99}$ on the out-of-sample set, while J_{mean} is used to monitor the average cost level. Computation time is also recorded for each method, including cost-matrix generation, uncertainty-set construction when applicable, and the final assignment optimization.

Evaluation of Robustness under Target Uncertainty

Figures 4–9 present box-plot summaries over the 300 Monte Carlo engagement geometries, using the evaluation metrics defined in (30) and (32). In each box plot, the box indicates the interquartile range from the 25th to the 75th percentile, the yellow line denotes the median, and the green triangle denotes the mean.

Figures 4–6 show the construction-set evaluation results. In Fig. 4, deterministic optimization gives the lowest mean cost because it optimizes the nominal PIP realization without accounting for uncertainty. The robust methods generally have slightly larger mean costs, which indicates the expected trade-off between nominal performance and robustness. Among the robust methods, SOCP-Sigma shows a lower mean cost than SOCP-Random and the exhaustive sample-based search.

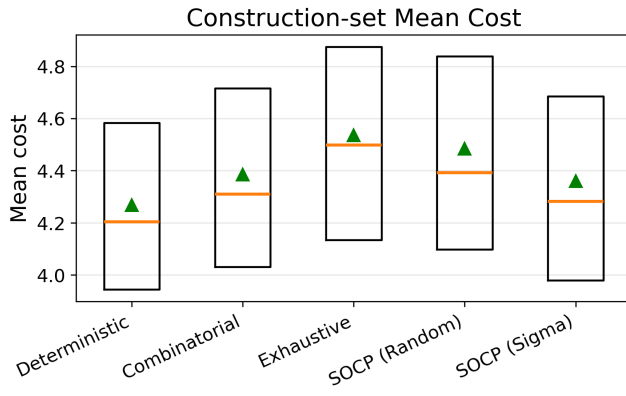


Figure 4. Construction-set mean assignment cost for the baseline methods and the proposed SOCP variants.

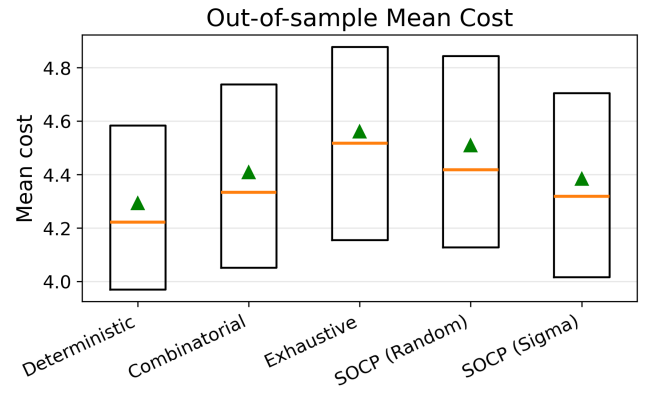


Figure 7. Out-of-sample mean assignment cost evaluated on independent Gaussian PIP realizations.

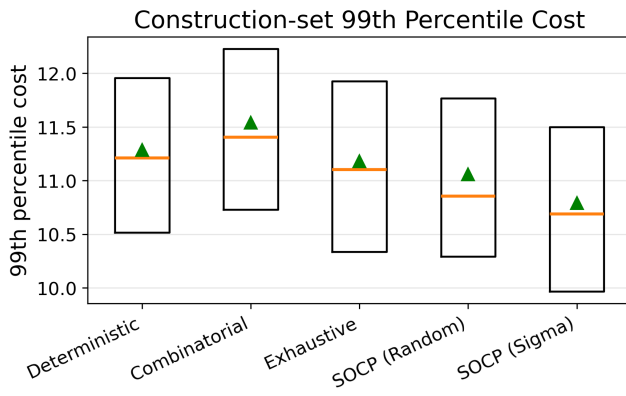


Figure 5. Construction-set 99th-percentile assignment cost, representing the upper-tail cost threshold exceeded by only 1% of the construction samples.

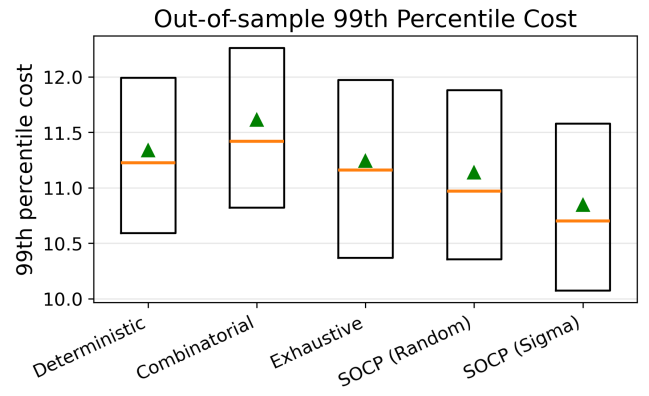


Figure 8. Out-of-sample 99th-percentile assignment cost evaluated on independent Gaussian PIP realizations.

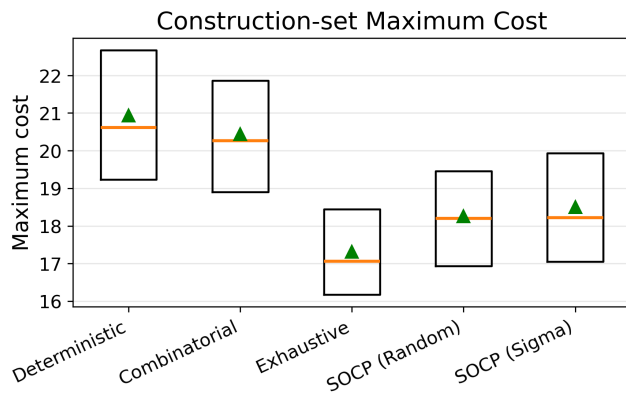


Figure 6. Construction-set maximum assignment cost over the PIP samples used for uncertainty-set construction.

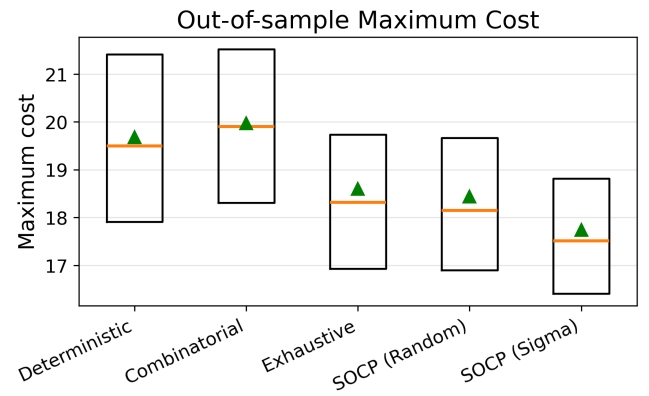


Figure 9. Out-of-sample maximum assignment cost evaluated on independent Gaussian PIP realizations.

Figure 5 compares the 99th-percentile construction-set costs. The deterministic and budgeted-uncertainty combinatorial methods show larger upper-tail costs than the proposed SOCP variants. SOCP-Sigma gives the lowest average 99th-percentile cost among the tested methods, indicating that the sigma-point ellipsoidal construction reduces high-percentile costs without requiring the large random sample set used by SOCP-Random.

Figure 6 shows the maximum cost over the construction samples. As expected, the exhaustive sample-based search gives the lowest construction-set maximum cost because

it directly minimizes the maximum realized cost over the same finite sample set. The SOCP variants are not optimized directly over individual samples, but over the ellipsoidal approximation in cost space, so their construction-set maximum costs are slightly higher than the exhaustive benchmark. Nevertheless, both SOCP variants reduce the construction-set maximum cost relative to deterministic optimization.

Figures 7–9 show the out-of-sample evaluation results on independent Gaussian PIP realizations. In Fig. 7, deterministic optimization again has the lowest mean cost, while the robust methods incur modestly larger mean costs.

This indicates a modest trade-off: the robust methods reduce large-cost outcomes, while their mean assignment costs are slightly higher than that of deterministic optimization.

Figure 8 compares the out-of-sample 99th-percentile cost. SOCP-Sigma gives the lowest value, followed by SOCP-Random. The budgeted-uncertainty combinatorial method gives a larger 99th-percentile cost than deterministic optimization in this experiment. This result suggests that the componentwise min-max uncertainty description used by the budgeted-uncertainty method is less effective than the ellipsoidal cost-space construction for this problem.

Figure 9 shows the maximum out-of-sample cost. SOCP-Sigma gives the lowest average maximum cost, while SOCP-Random also improves over deterministic optimization and the combinatorial baseline. The exhaustive sample-based search is no longer the best method on the independent test set, even though it performs best on the construction-set maximum cost. This is because the exhaustive benchmark is optimized over a finite construction sample, whereas the out-of-sample evaluation uses unseen PIP realizations.

Overall, the six figures show a consistent trend. Deterministic optimization gives the lowest mean cost, but its 99th-percentile and maximum costs are higher than those of the proposed SOCP variants. For the WTA problem considered in this study, the budgeted-uncertainty combinatorial method is less effective because it treats cost variations componentwise and does not account for the relationships among pairwise costs induced by common PIP perturbations and assignment constraints. The exhaustive sample-based search gives the lowest maximum cost on the construction set, but it is not the best method on the independent out-of-sample set. Among the tested methods, SOCP-Sigma provides the lowest out-of-sample 99th-percentile and maximum costs with only a modest increase in mean cost.

Runtime Analysis and Scalability

Table 2 summarizes the scalability results for engagement sizes ranging from 6-vs-4 to 15-vs-10. The SOCP-based methods consistently reduce the out-of-sample maximum cost relative to deterministic optimization and the Bertsimas-Sim baseline. In the 15-vs-10 case, SOCP-Random and SOCP-Sigma reduce the mean out-of-sample maximum cost from 42.67 to 32.14 and 32.01, respectively.

The exhaustive sample-based benchmark is not included in the scalability table for engagement sizes larger than 6-vs-4 because its computational cost grows combinatorially with the number of interceptors and targets. In the 6-vs-4 case, the feasible assignment set contains 1,560 assignments and can still be exhaustively evaluated. However, for larger engagement sizes, the number of feasible many-to-many assignments increases rapidly, making exhaustive evaluation over all feasible assignments and all sampled PIP realizations computationally impractical. Therefore, the scalability comparison focuses on Deterministic, Bertsimas-Sim, SOCP-Random, and SOCP-Sigma, which remain executable across the tested engagement sizes.

The main difference between the two SOCP variants is computational. SOCP-Random is dominated by random-sample cost generation and MVEE fitting over 20,000 sampled cost vectors, whereas SOCP-Sigma constructs the

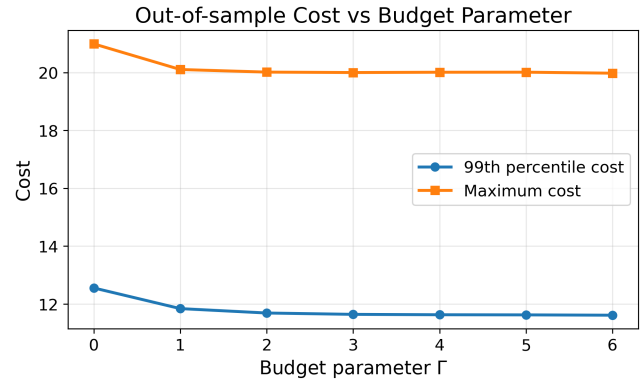


Figure 10. Out-of-sample 99th-percentile and maximum assignment costs for the budgeted-uncertainty combinatorial method under different values of the uncertainty budget Γ .

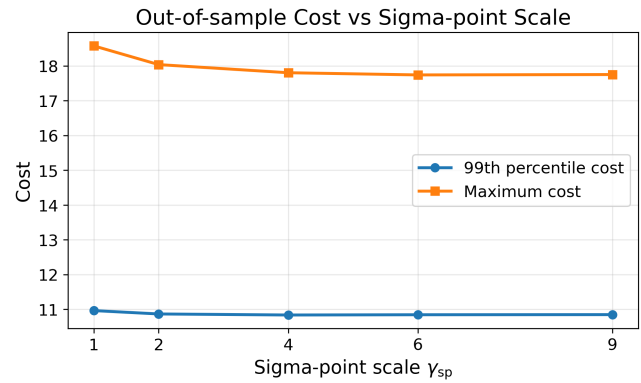


Figure 11. Out-of-sample 99th-percentile and maximum assignment costs for SOCP-Sigma under different sigma-point scale parameters γ_{sp} .

uncertainty set from only $6N + 1$ deterministic sigma-point realizations. As a result, SOCP-Sigma remains substantially faster across all tested engagement sizes. In the largest 15-vs-10 case, it requires 8.536 s on average, compared with 228.79 s for SOCP-Random, while producing a very similar out-of-sample maximum cost. These results indicate that the sigma-point construction provides the more scalable robustness-runtime trade-off for larger WTA instances.

Parameter Sensitivity

This subsection examines the sensitivity of the main algorithmic parameters used in the benchmark and proposed methods. Specifically, the uncertainty budget Γ in the budgeted-uncertainty combinatorial method, the sigma-point scale parameter γ_{sp} in SOCP-Sigma, and the MVEE diagonal loading parameter ϵ are varied while keeping the remaining simulation settings fixed.

Figure 10 shows the effect of the uncertainty budget Γ in the budgeted-uncertainty combinatorial method. Increasing Γ from 0 to 6 reduces the out-of-sample 99th-percentile and maximum costs, indicating that some protection against cost variation is beneficial. However, the improvement becomes small after $\Gamma = 4$, and $\Gamma = 6$ gives the lowest maximum cost among the tested values. Therefore, $\Gamma = 6$ is used in the main comparison.

For the present WTA formulation, values of Γ larger than 6 do not provide additional protection. This is because

Table 2. Scalability results for different engagement sizes. The table reports the mean out-of-sample maximum assignment cost and the mean end-to-end computation time over 10 Monte Carlo engagement geometries. For uncertainty-aware methods, the reported time includes uncertainty-set or uncertainty-range construction and assignment optimization.

Engagement size	Method	Out-of-sample max cost	Mean computation time [s]
6-vs-4	Deterministic	20.10	0.009
	Exhaustive	18.60	341.3
	Combinatorial	19.75	34.57
	SOCP-Random	18.14	68.82
	SOCP-Sigma	17.50	0.145
9-vs-6	Deterministic	28.67	0.011
	Combinatorial	27.72	74.96
	SOCP-Random	23.50	160.72
	SOCP-Sigma	23.14	0.515
12-vs-8	Deterministic	34.01	0.016
	Combinatorial	33.76	142.69
	SOCP-Random	28.04	172.98
	SOCP-Sigma	27.85	2.743
15-vs-10	Deterministic	42.67	0.024
	Combinatorial	42.48	236.93
	SOCP-Random	32.14	228.79
	SOCP-Sigma	32.01	8.536

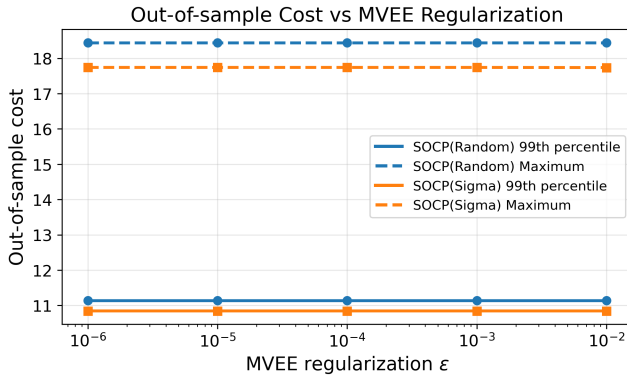


Figure 12. Out-of-sample 99th-percentile and maximum assignment costs for the SOCP-MVEE variants under different MVEE diagonal loading parameters ϵ .

each feasible assignment selects exactly one target for each interceptor, so the number of selected pairwise cost entries is always equal to the number of interceptors, $M = 6$. In the Bertsimas–Sim formulation, Γ limits how many selected cost entries can take their worst deviations. Once $\Gamma \geq M$, all selected entries can already be fully protected, and increasing Γ beyond M does not change the robust objective. This explains why the results are identical for $\Gamma > 6$.

Figure 11 shows the effect of the sigma-point scale parameter γ_{sp} in SOCP-Sigma. The parameter determines how far the deterministic sigma points are placed from the nominal PIP. In the implementation, the sigma points are located at $\bar{\mathbf{p}} \pm \sqrt{\gamma_{sp}}$ times the standard-deviation direction of the Gaussian PIP model. Thus, increasing γ_{sp} expands the PIP uncertainty region represented by the sigma points and generally makes the resulting cost-space ellipsoid more conservative.

The out-of-sample results show that the maximum cost decreases as γ_{sp} increases up to $\gamma_{sp} = 6$. The 99th-percentile

cost changes only slightly between $\gamma_{sp} = 4$ and $\gamma_{sp} = 6$, while increasing the scale further to $\gamma_{sp} = 9$ does not further reduce the maximum or 99th-percentile cost. Therefore, $\gamma_{sp} = 6$ is selected for the main comparison because it gives the lowest out-of-sample maximum cost among the tested values while maintaining nearly the same 99th-percentile cost as the best tested value.

Figure 12 examines the sensitivity of the proposed SOCP-MVEE formulations to the MVEE diagonal loading parameter ϵ . This parameter is introduced in (26) to ensure numerical regularity when the empirical cost covariance is low-rank or nearly low-rank, which is especially relevant for the sigma-point construction. The tested range spans $\epsilon = 10^{-6}$ to 10^{-2} .

Across the tested range, both the 99th-percentile and maximum out-of-sample costs vary only marginally for SOCP-Random and SOCP-Sigma. This suggests that, for the present simulation setting, the assignment-level performance is governed primarily by the dominant cost-variation directions captured by the construction samples, rather than by the precise value of the diagonal loading parameter. The role of ϵ is therefore mainly numerical: it stabilizes the MVEE matrix recovery and ensures a positive-definite ellipsoidal representation.

The diagonal loading term is also useful for the sigma-point construction because the number of sigma-point cost vectors may be smaller than the dimension of the assignment-cost space as the engagement size increases. In the three-dimensional setting, each cost vector lies in an $n_c = NM$ dimensional space, whereas the sigma-point construction uses $m_{sp} = 6N + 1$ PIP realizations. Hence,

$$\begin{aligned} \text{rank}(D_{sp}) &\leq m_{sp} = 6N + 1, \\ D_{sp} &= [d_1 \ d_2 \ \cdots \ d_{m_{sp}}] \in \mathbb{R}^{NM \times m_{sp}}. \end{aligned} \quad (33)$$

When $6N + 1 < NM$, the sigma-point cost matrix cannot span the full assignment-cost space. Even when $6N + 1 \geq$

NM , the sigma-point perturbations are deterministic and highly structured, so the resulting cost vectors may excite only a limited subset of cost-space directions. The associated covariance-like matrix can therefore be ill-conditioned. The diagonal loading term ϵI is used to stabilize the MVEE recovery.

Validation of the Assignment-Level Control-Effort Surrogate

To examine whether the assignment-level control-effort surrogate preserves the cost trends obtained from a more detailed engagement model, we performed an additional validation study using a PN-based three-dimensional reference guidance simulation. This analysis is not intended to replace the assignment surrogate with a full guidance simulation inside the optimization loop. Instead, it assesses whether the surrogate preserves the assignment-cost ordering produced by the PN-based three-dimensional reference model.

For each Monte Carlo engagement geometry, the missile and target initial states were generated using the same three-dimensional engagement model used in the main simulations. For each geometry, 300 Gaussian PIP perturbation samples were generated with $\sigma = 6.0$. The surrogate cost matrix was computed using the reduced-order assignment-level control-effort model, while the PN-based reference cost matrix was computed using a three-dimensional proportional navigation simulation with time step $\Delta t = 0.05$, final time 30.0, navigation gain $N_{PN} = 3.0$, acceleration limit $a_{\max} = 0.5$, first-order autopilot lag time constant $\tau_{PN} = 0.2$, and hit radius 0.2. This validation process was repeated for 300 independently generated engagement geometries.

The validation was conducted at the assignment level rather than only at the individual missile–target pair level. For each engagement geometry, 300 feasible many-to-many assignments were selected from the feasible assignment set. For a given assignment X , the surrogate and PN assignment costs were computed as

$$\begin{aligned} J_{\text{surrogate}}(X, \xi_k) &= \sum_{i=1}^N \sum_{j=1}^M C_{ij}^{\text{surrogate}}(\xi_k) X_{ij}, \\ J_{\text{PN}}(X, \xi_k) &= \sum_{i=1}^N \sum_{j=1}^M C_{ij}^{\text{PN}}(\xi_k) X_{ij}, \end{aligned} \quad (34)$$

where ξ_k denotes the k -th PIP perturbation sample. This produces a direct assignment-level comparison between the proposed surrogate and the PN simulation over multiple engagement geometries, assignments, and uncertainty samples.

Two complementary correlation diagnostics were then evaluated. First, a log-scale scatter plot was generated using the assignment-level pairs $(J_{\text{surrogate}}, J_{\text{PN}})$ pooled across cases, assignments, and PIP samples. The logarithmic scale was used because assignment costs span multiple orders of magnitude. Second, for each engagement geometry, the Spearman rank correlation coefficient ρ was computed between the surrogate assignment costs and PN assignment costs. This rank-correlation analysis measures whether

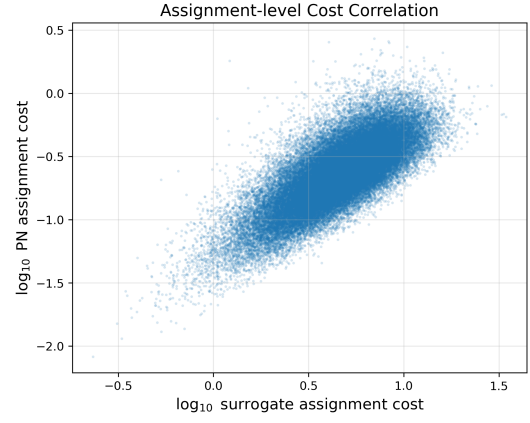


Figure 13. Assignment-level cost correlation between the proposed control-effort surrogate and the PN-based three-dimensional reference simulation. Each point represents a feasible assignment under one sampled PIP perturbation. Both axes are shown on a logarithmic scale because the assignment costs span multiple orders of magnitude.

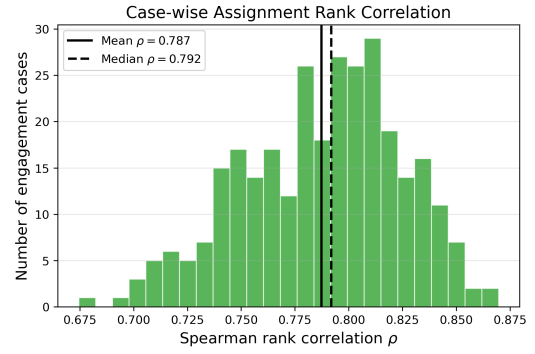


Figure 14. Case-wise Spearman rank correlation between surrogate assignment costs and PN-based reference assignment costs. Each value of ρ is computed over the selected feasible assignments and PIP samples for one engagement geometry.

the surrogate preserves the relative ordering of candidate assignments, which is the key property needed for WTA optimization.

Figure 13 shows the assignment-level cost relationship between the surrogate model and the PN simulation. The points follow a clear increasing trend on the logarithmic scale, indicating that assignments with larger surrogate costs also tend to have larger PN-simulation costs. This result supports the use of the reduced-order surrogate as an assignment-level cost model: although the surrogate does not reproduce every detail of the PN-based three-dimensional reference dynamics, it captures the dominant variation in assignment cost induced by PIP uncertainty.

Figure 14 further evaluates this relationship from a decision-making perspective. Because the WTA problem depends primarily on the relative ranking of feasible assignments, rank preservation is more important than exact pointwise cost matching. The distribution of Spearman rank correlations shows that the surrogate generally preserves the assignment ordering produced by the PN simulation across engagement geometries. Quantitatively, the mean and median case-wise Spearman rank correlations are 0.787 and

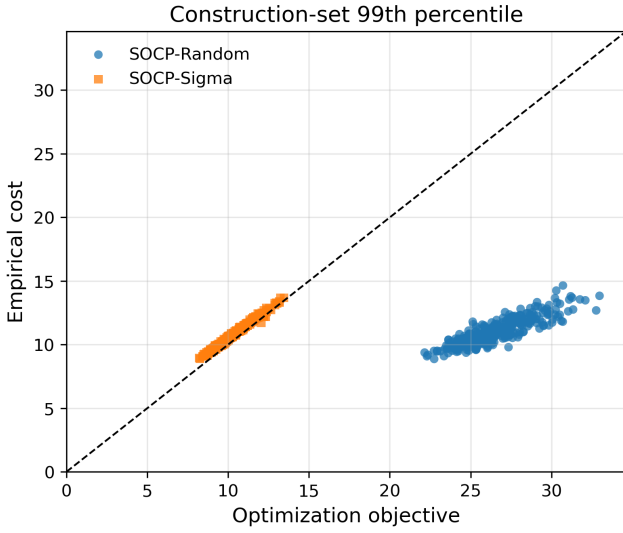


Figure 15. Case-wise comparison between the robust optimization objective and the construction-set 99th-percentile assignment cost. Each point corresponds to one Monte Carlo engagement geometry. The diagonal line denotes equality between the objective value and the empirical cost.

0.792, respectively, over the 300 engagement geometries. This indicates that the reduced-order surrogate preserves a substantial portion of the assignment-level ranking information produced by the PN-based reference simulation. Therefore, the reduced-order control-effort model provides a computationally efficient proxy for assignment evaluation while retaining the ranking information needed for robust WTA optimization.

These results support the modeling choice used in the main robust optimization framework. The PN-based three-dimensional reference simulation is more detailed than the assignment-level surrogate but remains too expensive to embed directly inside repeated uncertainty propagation and mixed-integer robust optimization. The proposed surrogate enables large-scale cost-space sampling and MVEE construction, while the validation results indicate that it remains consistent with the PN-based reference simulation at the assignment level.

Conservatism and Calibration of the Robust Objective

This subsection analyzes the conservatism and empirical calibration of the MVEE-based robust objective. Beyond comparing the realized assignment costs across methods, we examine whether the SOCP objective provides an empirical upper bound on the costs observed over the construction and out-of-sample evaluation sets. We also investigate how this relationship changes as the size of the uncertainty approximation is varied. For each Monte Carlo engagement geometry, the SOCP solution z^* gives the robust objective

$$J_{\text{rob}}(z^*) = c^T z^* + \|E^{-1/2} z^*\|_2, \quad (35)$$

which corresponds to the worst-case cost over the ellipsoidal uncertainty set used in (17). This value is compared with the empirical metrics $J_{0.99}(z^*; \mathcal{D})$ and $J_{\text{max}}(z^*; \mathcal{D})$ defined in (30) and (32), using both the construction set and the independent out-of-sample set.

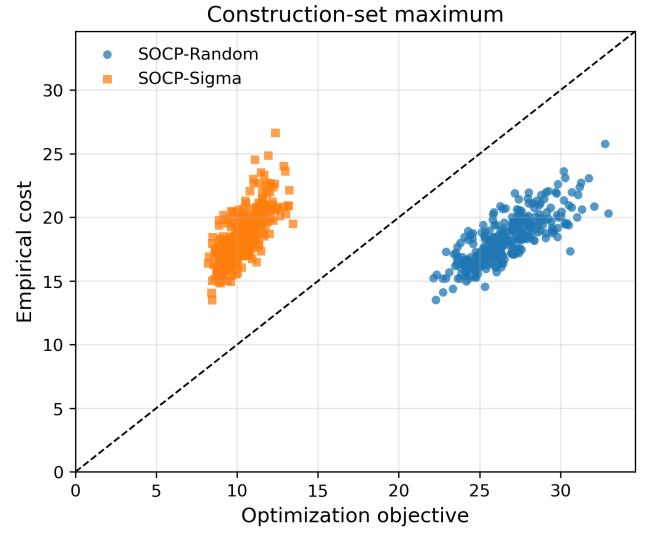


Figure 16. Case-wise comparison between the robust optimization objective and the construction-set maximum assignment cost. SOCP-Random remains below the equality line for all cases, indicating a conservative empirical upper bound over the construction samples.

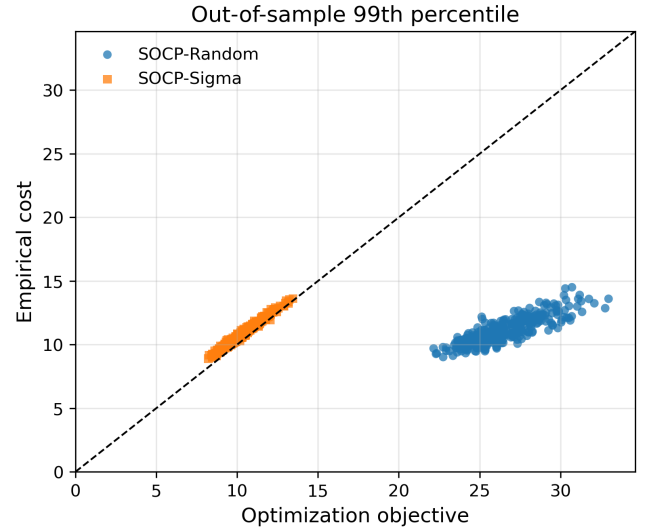


Figure 17. Case-wise comparison between the robust optimization objective and the out-of-sample 99th-percentile assignment cost. The SOCP-Sigma objective is closely calibrated to the high-probability cost region, whereas SOCP-Random gives a more conservative upper bound.

Figures 15–18 show the resulting case-wise calibration plots. In each figure, the horizontal axis is the robust optimization objective $J_{\text{rob}}(z^*)$, and the vertical axis is the corresponding empirical assignment cost. The diagonal line denotes equality between the robust objective and the empirical cost. Points below the diagonal therefore indicate cases in which the robust objective upper-bounds the empirical cost, whereas points above the diagonal indicate that the empirical cost exceeds the objective value.

For SOCP-Random, all 300 Monte Carlo cases lie below the equality line for both the construction-set and out-of-sample maximum costs. Quantitatively, the mean robust objective is 26.63, while the mean construction-set maximum and out-of-sample maximum costs are 18.26 and

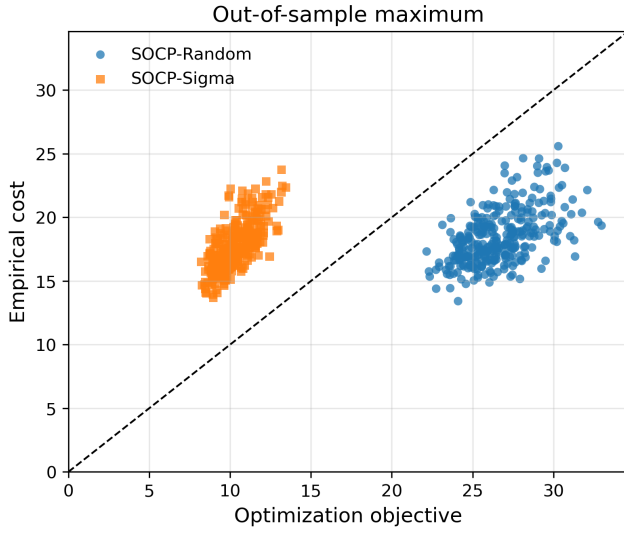


Figure 18. Case-wise comparison between the robust optimization objective and the out-of-sample maximum assignment cost. SOCP-Random upper-bounds the empirical maximum cost in all Monte Carlo cases, while SOCP-Sigma does not upper-bound the maximum tail realization and is therefore interpreted as a less conservative high-probability approximation.

18.44, respectively. Thus, the random-sample MVEE yields a consistently conservative upper bound on the empirical worst-case cost. This behavior is expected because the uncertainty set is constructed from a large number of random PIP realizations and therefore produces a broader cost-space envelope.

For SOCP-Sigma, the relationship is different. The robust objective is close to the empirical 99th-percentile costs: the mean objective is 10.33, while the mean construction-set and out-of-sample 99th-percentile costs are 10.79 and 10.84, respectively. However, the same objective does not upper-bound the empirical maximum costs, whose means are 18.50 and 17.74 on the construction and out-of-sample sets. Thus, the sigma-point construction should not be interpreted as a worst-case upper-bound approximation of the full Gaussian tail. Rather, it provides a less conservative approximation calibrated near the high-probability cost region. This explains why SOCP-Sigma gives favorable 99th-percentile performance while maintaining a substantially lower computational cost than SOCP-Random.

The construction-set plots also show an approximately linear relationship between $J_{\text{rob}}(z^*)$ and the empirical costs for each SOCP variant. No isolated outlier or distinct cluster is observed. This suggests that the observed conservatism is mainly governed by the overall scale of the ellipsoidal uncertainty set rather than by a specific case-wise ellipsoid geometry that produces exceptionally large or small conservatism. In other words, the MVEE over-approximation acts primarily as a global calibration of the robust objective: the random-sampling construction gives a conservative empirical upper bound, whereas the sigma-point construction gives a tighter, less conservative objective that remains meaningful for high-percentile cost reduction.

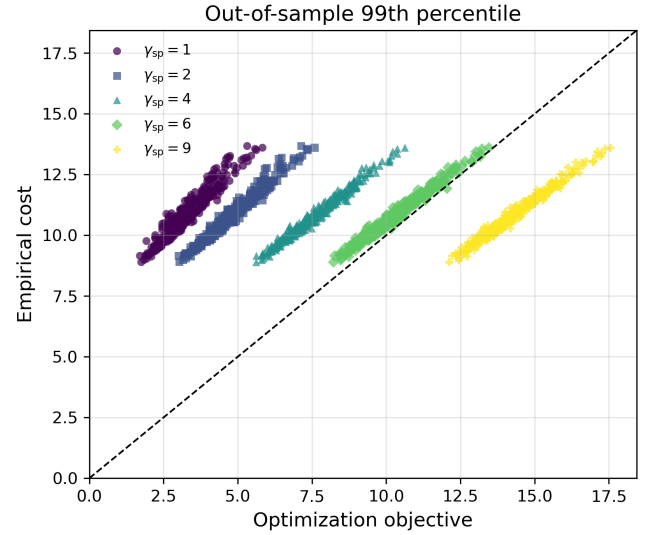


Figure 19. Case-wise comparison between the SOCP-Sigma objective and the out-of-sample 99th-percentile assignment cost for different sigma-point scale parameters γ_{sp} . Increasing γ_{sp} enlarges the sigma-point uncertainty approximation and shifts the robust objective toward more conservative values.

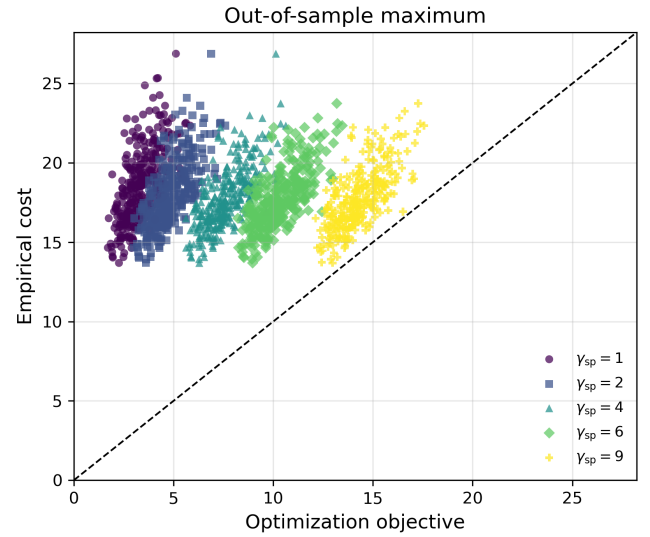


Figure 20. Case-wise comparison between the SOCP-Sigma objective and the out-of-sample maximum assignment cost for different sigma-point scale parameters γ_{sp} . Larger values of γ_{sp} increase objective conservatism, but the empirical maximum cost does not decrease proportionally, indicating saturation of the assignment-level performance gain.

The effect of the sigma-point scale parameter is further examined in Figs. 19 and 20. These figures compare the SOCP-Sigma objective with the out-of-sample 99th-percentile and maximum costs while varying γ_{sp} . As γ_{sp} increases, the deterministic sigma points are placed farther from the nominal PIP, which enlarges the corresponding cost-space ellipsoid. Consequently, the robust objective shifts to larger values and the points move progressively to the right of the plots.

This trend shows that γ_{sp} acts as a direct conservatism-control parameter for the sigma-point approximation. For small values of γ_{sp} , the robust objective is smaller than the empirical out-of-sample costs in many cases,

indicating that the resulting ellipsoid is not sufficiently conservative to upper-bound the observed tail realizations. Increasing γ_{sp} reduces this underestimation by increasing the objective value. Around the nominal value $\gamma_{sp} = 6$, the robust objective becomes closely calibrated to the out-of-sample 99th-percentile cost, while still remaining below the empirical maximum cost. Further increasing the scale to $\gamma_{sp} = 9$ makes the objective more conservative relative to the 99th-percentile cost, but it does not produce a comparable improvement in the empirical maximum cost. This indicates that enlarging the sigma-point uncertainty set primarily increases objective conservatism, whereas the assignment-level empirical performance eventually saturates.

Discussion of Modeling and Evaluation Limitations

Several limitations should be kept in mind when interpreting the reported results. First, the robustness achieved by the proposed method is conditional on the assumed Gaussian PIP model and on its ellipsoidal approximation in cost space. The method protects against modeled PIP-induced cost variations within the constructed uncertainty set, but it does not provide guarantees against structural model mismatch, unmodeled nonlinear engagement effects, or uncertainty sources that lie outside the assumed PIP model.

Second, the MVEE should be interpreted as a tractable outer approximation of the sampled cost cloud rather than as an exact probabilistic confidence region. Its conservatism depends on the geometry of the sampled cost vectors, the sampling strategy, and the regularization used in the MVEE recovery. The calibration results show this distinction clearly: the random-sampling construction behaves as a conservative empirical upper-bound approximation, whereas the sigma-point construction is better interpreted as a lower-cost approximation calibrated near high-probability cost regions.

Third, the sigma-point construction substantially reduces computational cost, but it does not fully represent the tails of the continuous PIP distribution. Its performance depends on whether the deterministic sigma-point perturbations sufficiently excite the dominant assignment-cost directions. The sigma-point variant should therefore be viewed as a computationally efficient robustness approximation, not as a worst-case guarantee over the full Gaussian uncertainty model.

Fourth, the PN-based validation supports the assignment-level ranking consistency of the reduced-order surrogate, but it does not make the surrogate a full-fidelity guidance model. The validation is intended to assess whether the surrogate preserves cost trends relevant to WTA decisions, rather than to reproduce all nonlinear interceptor dynamics and terminal guidance effects.

Finally, the brute-force reference is exhaustive only over the sampled evaluation set. It is therefore best interpreted as an exhaustive evaluation-set benchmark, not as the exact robust optimum over a continuous uncertainty set. Future work should include broader scaling experiments, alternative uncertainty descriptions such as polyhedral or other data-driven sets, numerically conditioned MVEE construction methods, and periodically replanned or

dynamic WTA formulations in which both target information and assignment decisions evolve over time.

Conclusion and Future Work

This paper presented a robust WTA framework for many-to-many interceptor–target engagements under uncertainty in predicted intercept points. Pairwise assignment costs were generated using an assignment-level reduced-order planar linear optimal-control surrogate, and PIP uncertainty was propagated into the assignment-cost space through either random samples or deterministic sigma points. The resulting cost vectors were compressed into an ellipsoidal uncertainty set using the MVEE, leading to a robust counterpart of the deterministic binary assignment problem that can be solved as an integer SOCP. The main contribution is therefore a modular robust WTA pipeline that combines control-effort-based pairwise cost modeling, PIP-to-cost uncertainty propagation, ellipsoidal cost-space compression, and mixed-integer conic assignment optimization.

The Monte Carlo results indicate that the proposed framework improves upper-tail and worst-case behavior relative to deterministic optimization and the budgeted-uncertainty combinatorial baseline in the 6-vs-4 engagement scenarios studied. Relative to the exhaustive sample-based benchmark used for empirical comparison, both SOCP variants substantially reduce the observed worst-case gap. The random-sampling variant provides more conservative empirical protection, whereas the sigma-point variant achieves comparable high-percentile robustness with a much lower uncertainty-set construction cost. These results demonstrate a practical robustness–runtime trade-off between empirical conservatism and computational efficiency.

The additional PN-based validation showed that the reduced-order surrogate preserves assignment-level cost trends and rank ordering relative to a PN-based three-dimensional reference guidance simulation, supporting its use as a computational proxy inside the robust WTA loop. Nevertheless, the reported robustness remains conditional on the assumed PIP uncertainty model and its ellipsoidal approximation in cost space; structural model mismatch, richer nonlinear engagement effects, and broader uncertainty classes remain outside the guarantees of the present formulation.

The main practical value of the proposed framework lies in providing a modular and computationally explicit robustness–runtime trade-off for assignment under modeled PIP uncertainty.

Future work will therefore focus on five directions: integrating higher-fidelity or precomputed pairwise cost generators into the same robust-assignment pipeline; investigating alternative uncertainty representations, such as polyhedral or other data-driven sets, to reduce ellipsoidal conservatism; improving the numerical conditioning of MVEE construction, for example through normalized or whitened cost-space preprocessing; extending the framework to dynamic or periodically replanned WTA problems with time-varying uncertainty; and developing theoretical bounds that explain the conservatism of the proposed approximation independently of Monte Carlo evaluation.

Acknowledgements

During the preparation of this work, the authors used ChatGPT by OpenAI to check grammar, spelling, phrasing, and equation formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Statements and Declarations

Ethical Considerations

Not applicable. This study did not involve human participants or animals.

Consent to Participate

Not applicable. This study did not involve human participants.

Consent for Publication

Not applicable. This study does not include identifiable personal data or images.

Declaration of Conflicting Interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding Statement

This work was supported in part by the Theater Defense Research Center funded by the Defense Acquisition Program Administration (DAPA) under Grant No. UD240002SD, and in part by the Korea Research Institute for Defense Technology Planning and Advancement (KRIT) under Grant No. 20-105-E00-005 (KRIT-CT-23-010) through the VTOL Technology Research Center for Defense Applications, funded by Korea government (DAPA). Equal support was received from both funding sources.

Data Availability Statement

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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